

Dataset Description – Global Property Purchase Decision Dataset

1. Overview and Background

The Global Property Purchase Decision Dataset is an extensive compilation of property transactions and buyer profiles sourced from over 20 countries and prominent global cities. Created to support academic research, machine learning applications, and market trend analysis, this dataset aims to reveal the key factors that affect individuals' decisions to purchase real estate around the world. Containing 25,000 records, it reflects a wide variety of international market contexts, property classifications, financial profiles, and subjective assessments, enabling comprehensive modeling and insightful evaluation.

2. Dataset Composition

- **Total Records:** 25,000 individual property transactions or purchase decisions
- **Geographic Coverage:** Includes data from more than 20 countries, encompassing both nationwide and major metropolitan regions
- **Features:** Numerous fields covering property attributes, financial information, and buyer perceptions
- **Target Variable:** Decision — a binary indicator representing whether the buyer chose to purchase (1) or not (0)

3. Features Explained

Property Characteristics

- **Location:** Specifies the country and major city of the property, facilitating geographical analysis and segmentation
- **Size Details:** Contains measurements such as square footage, number of bedrooms, or rooms to describe the property's dimensions
- **Price Details:** Includes the sale price or estimated market value, supporting affordability and pricing studies
- **Property Type:** Classifies real estate into categories like apartment, house, villa, condominium, or commercial units

- **Available Amenities:** Lists included features such as parking, swimming pools, gyms, security services, gardens, and others

Financial Attributes

- **Income Information:** Captures buyer salary data to analyze purchasing power and loan eligibility
- **Loan Information:** Details about home loan approvals including amounts, terms, and lender types
- **EMI Amounts:** Monthly installment payments agreed upon by the buyer for any housing loans
- **Expense Breakdown:** Covers additional costs associated with the property, such as taxes, insurance, maintenance, and registration fees

Buyer Perceptions

- **Satisfaction Ratings:** Scores reflecting buyer satisfaction with the property and the purchasing experience
- **Neighborhood Evaluations:** Assesses quality of local environment, safety, and access to services
- **Connectivity Ratings:** Measures ease of transportation, availability of public transit, and commute convenience

Target Variable

- **Decision:** The focus variable indicating if the buyer proceeded with the purchase (1) or not (0), which is suitable for binary classification modeling tasks.